

my_pwm_6ch IP Register Description (English)

1. Register Summary

The **my_pwm_6ch** IP provides eight 32-bit registers through an AXI4-Lite slave interface. All registers are 4-byte aligned and reset to **0x00000000**.

Offset	Register	Access	Initial Value	Description
0x00	SLV_REG0 (PWM_DUTY1)	R/W	0x00000000	PWM high-time count for servo1
0x04	SLV_REG1 (PWM_DUTY2)	R/W	0x00000000	PWM high-time count for servo2
0x08	SLV_REG2 (PWM_DUTY3)	R/W	0x00000000	PWM high-time count for servo3
0x0C	SLV_REG3 (PWM_DUTY4)	R/W	0x00000000	PWM high-time count for servo4
0x10	SLV_REG4 (PWM_DUTY5)	R/W	0x00000000	PWM high-time count for servo5
0x14	SLV_REG5 (PWM_DUTY6)	R/W	0x00000000	PWM high-time count for servo6
0x18	SLV_REG6 (RESERVED0)	R/W	0x00000000	Reserved register not connected to current user logic
0x1C	SLV_REG7 (RESERVED1)	R/W	0x00000000	Reserved register not connected to current user logic

2. Register Description

2.1 SLV_REG0 - PWM Duty Register 1

Item	Value
Bit	31:0
Field	DUTY1[31:0]
Read/Write	R/W
Initial Value	0x00000000

SLV_REG0 controls the PWM high interval of the **servo1** output. The internal PWM logic drives **servo1** high while **counter** < **DUTY1**.

- Bits **31:0** - **DUTY1**: PWM duty count value The value written to this field defines the number of counts for which **servo1** remains high within one PWM frame. In the current implementation, **PERIOD** = **2,000,000**, which corresponds to an approximately 20 ms frame when **S_AXI_ACLK** = **100 MHz**, as indicated by the source comment. Under the same assumption, one count equals 10 ns, and typical servo pulse widths are 100,000 counts for 1.0 ms, 150,000 counts for 1.5 ms, and 200,000 counts for 2.0 ms.

2.2 **SLV_REG1** - PWM Duty Register 2

Item	Value
Bit	31:0
Field	DUTY2[31:0]
Read/Write	R/W
Initial Value	0x00000000

SLV_REG1 controls the PWM high interval of the **servo2** output. The internal PWM logic drives **servo2** high while **counter** < **DUTY2**.

- Bits **31:0** - **DUTY2**: PWM duty count value The value written to this field defines the number of counts for which **servo2** remains high within one PWM frame. The count unit and time conversion are identical to **SLV_REG0**.

2.3 **SLV_REG2** - PWM Duty Register 3

Item	Value
Bit	31:0
Field	DUTY3[31:0]
Read/Write	R/W
Initial Value	0x00000000

SLV_REG2 controls the PWM high interval of the **servo3** output. The internal PWM logic drives **servo3** high while **counter** < **DUTY3**.

- Bits **31:0** - **DUTY3**: PWM duty count value The value written to this field defines the number of counts for which **servo3** remains high within one PWM frame. The count unit and time conversion are identical to **SLV_REG0**.

2.4 **SLV_REG3** - PWM Duty Register 4

Item	Value
Bit	31:0

Item	Value
Field	DUTY4[31:0]
Read/Write	R/W
Initial Value	0x00000000

SLV_REG3 controls the PWM high interval of the **servo4** output. The internal PWM logic drives **servo4** high while **counter** < **DUTY4**.

- Bits **31:0** - **DUTY4**: PWM duty count value The value written to this field defines the number of counts for which **servo4** remains high within one PWM frame. The count unit and time conversion are identical to **SLV_REG0**.

2.5 SLV_REG4 - PWM Duty Register 5

Item	Value
Bit	31:0
Field	DUTY5[31:0]
Read/Write	R/W
Initial Value	0x00000000

SLV_REG4 controls the PWM high interval of the **servo5** output. The internal PWM logic drives **servo5** high while **counter** < **DUTY5**.

- Bits **31:0** - **DUTY5**: PWM duty count value The value written to this field defines the number of counts for which **servo5** remains high within one PWM frame. The count unit and time conversion are identical to **SLV_REG0**.

2.6 SLV_REG5 - PWM Duty Register 6

Item	Value
Bit	31:0
Field	DUTY6[31:0]
Read/Write	R/W
Initial Value	0x00000000

SLV_REG5 controls the PWM high interval of the **servo6** output. The internal PWM logic drives **servo6** high while **counter** < **DUTY6**.

- Bits **31:0** - **DUTY6**: PWM duty count value The value written to this field defines the number of counts for which **servo6** remains high within one PWM frame. The count unit and time conversion are identical to **SLV_REG0**.

2.7 SLV_REG6 - Reserved Register 0

Item	Value
Bit	31:0
Field	RESERVED0[31:0]
Read/Write	R/W
Initial Value	0x00000000

SLV_REG6 exists in the AXI address space but is not connected to the current `pwm_servo_6ch` user logic.

- Bits 31:0 - **RESERVED0**: Reserved field In the current implementation, this field does not affect any PWM output or internal control signal. For forward compatibility, software should write 0x00000000 unless a future user extension defines another purpose.

2.8 SLV_REG7 - Reserved Register 1

Item	Value
Bit	31:0
Field	RESERVED1[31:0]
Read/Write	R/W
Initial Value	0x00000000

SLV_REG7 exists in the AXI address space but is not connected to the current `pwm_servo_6ch` user logic.

- Bits 31:0 - **RESERVED1**: Reserved field In the current implementation, this field does not affect any PWM output or internal control signal. For forward compatibility, software should write 0x00000000 unless a future user extension defines another purpose.

3. Operation Notes

- SLV_REG0 through SLV_REG5 map one-to-one to `servo1` through `servo6`.
- All registers support software read-back, and a read returns the last written value.
- During reset (`S_AXI_ARESETN = 0`), all registers are cleared to 0, so all PWM outputs start inactive.
- If a `DUTYx` value is greater than `PERIOD`, the internal counter never reaches that value before reset, so the corresponding output can remain high for almost the entire frame.
- The timing examples in this document are based on the source definition `PERIOD = 2,000,000` and the source comment `20 ms @100 MHz`. If the actual system clock differs, the pulse width scales proportionally.